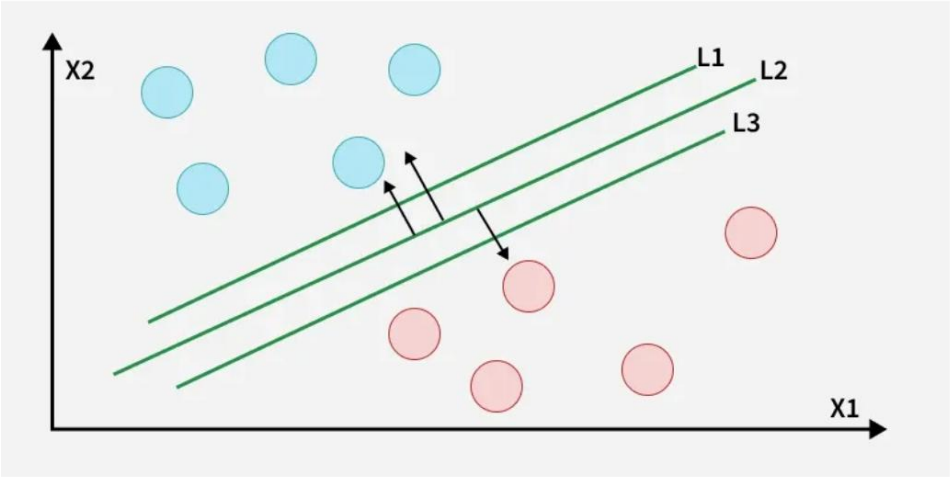
1.Support Vector Machine (SVM):

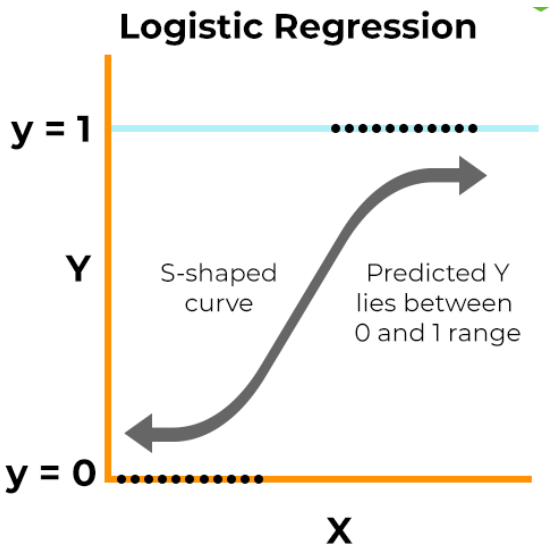
Support Vector Mahine is a supervised machine learning algorithm that is used to solve classification and regression problems. SVM works by drawing a clear boundary called hyperplane, between different classes in the data. The idea is to separate the data points in such a way that each class lies on one side of the boundary. It finds the best possible line. It does this by maximizing the distance between the boundary and the closest data points from each class. A larger margin means the model is more confident about its classification and is likely to perform better on new, unseen data.



Multiple hyperplanes separate the data from two classes

2.Logistic Regression:

Logistic Regression is a fundamental machine learning algorithm used for binary classification problems. It is used when we want to make decisions between two outcomes, like yes or no, 0 or 1, spam or not spam. Instead of directly giving a fixed answer, it first calculates the probability of the outcome. To do this, it uses something called the sigmoid function, which converts any value into a number between 0 and 1. This creates an S-shaped curve instead of a straight line. Based on this probability, the model decides which category the data belongs to.



3.Naive bayes:

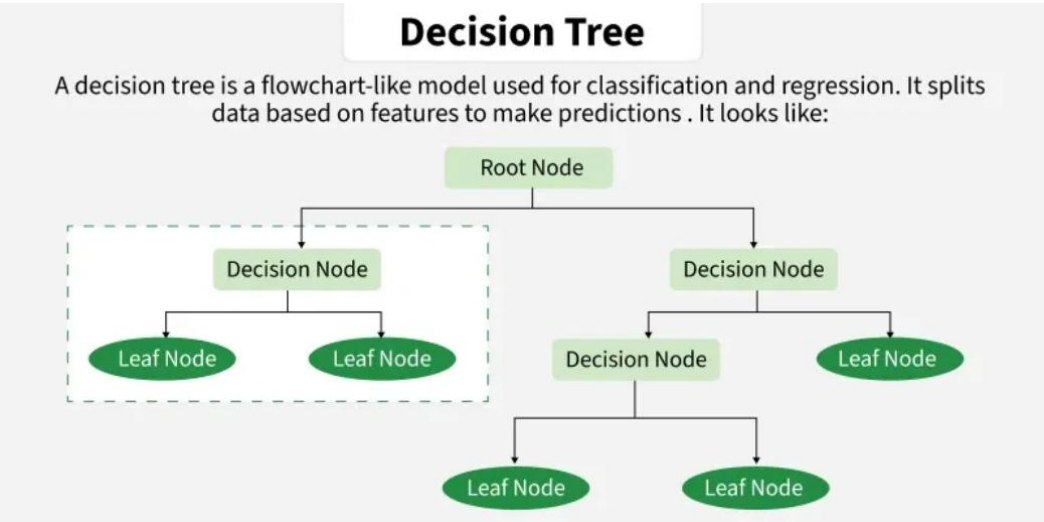
Naive bayes is a classification machine learning algorithm it is used to predict the category of a data point using probability. It works by assuming that all features are independent of each other. It calculates probabilities for each class, and picks the one with the highest probability. The main idea behind this classifier is to use Bayes' Theorem to classify data based on the probabilities of different classes. It is used mostly in high-dimensional text classification

Formula : $$P(C|X) = \frac{P(X|C) \cdot P(C)}{P(X)}$$

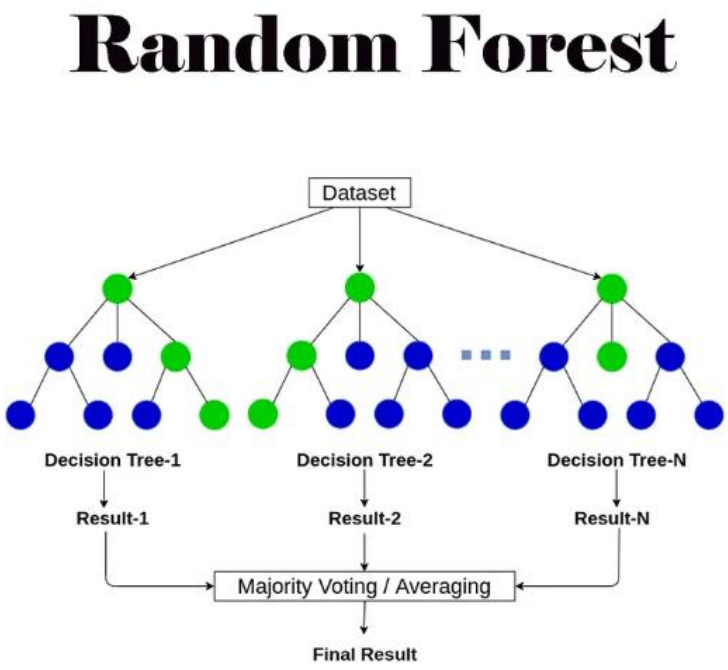
- P(C | X) → Probability of class *C* given the data X
- P(X | C) → Probability of data X given class C
- P(C) → Probability of class C
- P(X) → Probability of the data

4.Decision tree:

A Decision Tree is a supervised learning algorithm used for both classification and regression problem. It works like a flowchart or a series of decision-making steps. It splits the data into smaller parts based on conditions. It is like breaking down a problem into smaller parts, that makes it easy-to-understand the problem.



5.Random Forest:

Random Forest is an ensemble learning technique used for both classification and regression problems. Random Forest is an advanced version of decision trees, instead of relying on just one tree, it builds multiple decision trees and combines their results.

Each tree is trained on a different subset of the data, and combine their result by 'voting' for classification and calculates 'average' for regression.